Day 8 — JavaScript Fundamentals

Answers / Practice Programs

Covers the supplied task topics: variables, data types, operators, conditions, loops, functions, arrays, objects, and basic problem solving.

1. Variables: var, let and const

```
var name = "Shweta";
let age = 21;
const course = "Computer Science";

console.log("Name:", name);
console.log("Age:", age);
console.log("Course:", course);
```

var and let can be reassigned; const cannot be reassigned.

2. Data Types

```
let studentName = "Shweta"; // string
let marks = 85;              // number
let isPassed = true;         // boolean
let result;                  // undefined

console.log(typeof studentName);
console.log(typeof marks);
console.log(typeof isPassed);
console.log(typeof result);
```

typeof displays the type of a value.

3. Operators

```
let a = 20;
let b = 10;

console.log("Addition:", a + b);        // 30
console.log("Subtraction:", a - b);     // 10
console.log("Multiplication:", a * b);  // 200
console.log("Division:", a / b);       // 2
console.log("Remainder:", a % b);       // 0
```

Arithmetic operators perform calculations.

4. Conditional Statements

```
let marks = 75;

if (marks >= 90) {
    console.log("Grade A");
} else if (marks >= 75) {
    console.log("Grade B");
} else if (marks >= 50) {
    console.log("Grade C");
} else {
    console.log("Fail");
}

// Output: Grade B
```

if / else if / else selects a block based on a condition.

5. Even or Odd

```
let number = 7;

if (number % 2 === 0) {
    console.log("Even");
} else {
    console.log("Odd");
}

// Output: Odd
```

A number divisible by 2 with remainder 0 is even.

6. For Loop

```
for (let i = 1; i <= 5; i++) {
```

```
    console.log(i);
}

// Output: 1 2 3 4 5
```

A for loop repeats code while its condition is true.

7. While Loop

```
let count = 1;

while (count <= 5) {
    console.log(count);
    count++;
}

// Output: 1 2 3 4 5
```

A while loop checks its condition before each iteration.

8. Functions for Basic Calculations

```
function add(a, b) {
    return a + b;
}
function subtract(a, b) {
    return a - b;
}
function multiply(a, b) {
    return a * b;
}
function divide(a, b) {
    if (b === 0) return "Cannot divide by zero";
    return a / b;
}

console.log(add(10, 5));       // 15
console.log(subtract(10, 5));  // 5
console.log(multiply(10, 5));  // 50
console.log(divide(10, 5));    // 2
```

Functions are reusable code that can accept parameters and return results.

9. Array Operations

```
let fruits = ["Apple", "Mango", "Banana"];

console.log(fruits[0]);    // Apple
console.log(fruits.length); // 3

fruits.push("Orange");     // Add an item
console.log(fruits);

fruits.pop();              // Remove last item
console.log(fruits);
```

Arrays store ordered values. Indexing starts at 0.

10. Objects and Properties

```
let student = {
    name: "Shweta",
    age: 21,
    course: "Computer Science",
    marks: 85
};

console.log(student.name);
console.log(student["course"]);

student.marks = 90; // Update property
student.city = "Pune"; // Add property

console.log(student.marks);
console.log(student.city);
```

Objects store data as key-value pairs.

11. Sum of Numbers from 1 to 10

```
let sum = 0;

for (let i = 1; i <= 10; i++) {
    sum += i;
}
```

```
console.log("Sum:", sum);

// Output: Sum: 55
```

The loop adds each number to the running total.

12. Largest Number in an Array

```
let numbers = [12, 45, 23, 67, 34];
let largest = numbers[0];

for (let i = 1; i < numbers.length; i++) {
    if (numbers[i] > largest) {
        largest = numbers[i];
    }
}
console.log("Largest number:", largest);

// Output: Largest number: 67
```

Each array value is compared with the current largest value.

13. Multiplication Table

```
let tableNumber = 5;

for (let i = 1; i <= 10; i++) {
    console.log(tableNumber + " x " + i + " = " + (tableNumber * i));
}
```

Prints the multiplication table from 1 to 10.

14. Positive, Negative or Zero

```
let number = -5;

if (number > 0) {
    console.log("Positive");
} else if (number < 0) {
    console.log("Negative");
} else {
    console.log("Zero");
}

// Output: Negative
```

Conditions identify whether a number is positive, negative, or zero.

15. Function to Find Square

```
function square(number) {
    return number * number;
}
console.log(square(6));

// Output: 36
```

The function returns the number multiplied by itself.

How to Run

Save the code in **day8.js**. Run with Node.js using **node day8.js**, or add **<script src="day8.js"></script>** before the closing body tag of an HTML file and check the browser Console.